Unit 1 Practice Test - Version 6

Name: _____ Date: _____ 20 marks | 35 minutes

Circle one answer for each question. Each question is worth 1 mark. Calculator permitted; formal differentiation rules are not required.

1. Which relation is a function? [1]

A. {(-1,2),(0,2),(1,4)}
B. {(0,1),(0,2),(1,3)}
C. {(2,1),(3,2),(3,4)}
D. {(-2,0),(-2,0),(-2,1)}

2. What is the domain of f(x)=sqrt(x+2)/(x-1)? [1]

A. [-2,infinity)
B. (-2,1) U (1,infinity)
C. [-2,1) U (1,infinity)
D. (-infinity,1) U (1,infinity)

3. What is the range of g(x)=2|x-3|-5? [1]

A. (-infinity,-5]
B. [-5,infinity)
C. (-5,infinity)
D. all real numbers

4. Which interval represents -1<x<=4? [1]

A. [-1,4]
B. (-1,4)
C. [-1,4)
D. (-1,4]

5. If p(x)=2x+1 for x<3 and p(x)=x^2-4 for x>=3, what is p(3)? [1]

A. 3
B. 7
C. 5
D. 9

6. If p(x)=3x+2 for x<1 and p(x)=kx-1 for x>=1, which k makes p continuous at x=1? [1]

A. 4
B. 5
C. 3
D. 6

7. Which function is odd? [1]

A. x^4+2
B. x^5+2x
C. x^3+1
D. x^2+x

8. If (-6,4) lies on y=f(x), which point lies on h(x)=2f(-3(x+2))-1? [1]

A. (0,7)
B. (-4,7)
C. (0,-9)
D. (-4,-9)

9. Relative to y=f(x), what is the horizontal change in y=f(4(x-2))? [1]

A. Stretch by 4 and right 2.
B. Compress by factor 1/4 and right 2.
C. Compress by factor 1/4 and left 2.
D. Stretch by 4 and left 2.

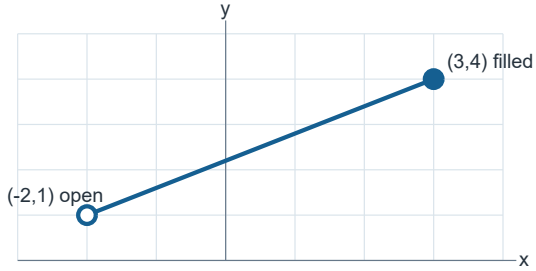
10. If f(x)=4x-7, what is f^(-1)(x)? [1]

A. 4x+7
B. (x-7)/4
C. (x+7)/4
D. 1/(4x-7)

11. If q(x)=-(x-2)^2+5 is restricted to x>=2, what is q^(-1)(x)? [1]

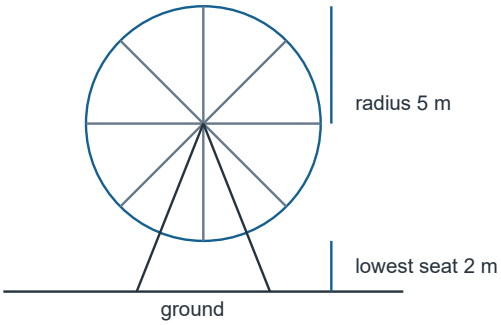
A. 2-sqrt(5-x)
B. 2+sqrt(x-5)
C. -2+sqrt(5-x)
D. 2+sqrt(5-x)

12. The graph is a line segment from an open point (-2,1) to a filled point (3,4). What is its domain? [1]



A. [-2,3]
B. (-2,3]
C. [-2,3)
D. (1,4]

Questions 13-14: The ride has radius 5 m, its lowest seat is 2 m above the ground, and one rotation takes 300 s. The rider starts at the bottom at t=0.



13. What is the greatest seat height? [1]

A. 12 m
B. 10 m
C. 7 m
D. 5 m

14. What is the seat height after 75 s? [1]

A. 2 m
B. 12 m
C. 7 m
D. 5 m

15. A function has `f(1)=2` and `f(4)=14`. What is its average rate of change on `[1,4]`? [1]

A. 3
B. 4
C. 12
D. 16

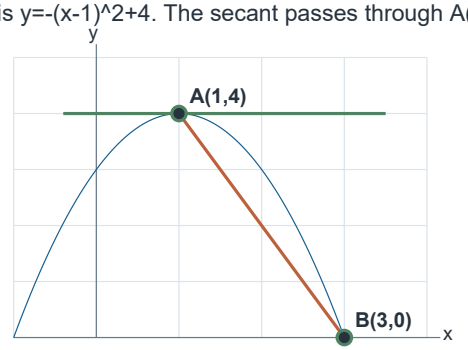
16. What is the average rate of change of `f(x)=-x^2+10x` on `[1,3]`? [1]

A. 4
B. 8
C. 6
D. 12

17. Using the centred interval `[1.9,2.1]`, estimate the instantaneous rate of change of `f(x)=x^2-6x` at `x=2`. [1]

A. 2
B. -2
C. -4
D. 0

Questions 18-19: The curve is y=-(x-1)^2+4. The secant passes through A(1,4) and B(3,0); the tangent at A is horizontal.



18. What is the slope of the secant AB? [1]

A. 0
B. 2
C. -2
D. -4

19. What is the instantaneous rate of change at A? [1]

A. 0
B. 2
C. -2
D. 4

20. What is the solution of `|2x-3|<=5` in interval notation? [1]

A. `(-1,4)`
B. `[-4,1]`
C. `(-4,1]`
D. `[-1,4]`